



Chemotactic self-organization captures the dynamics of mammalian hair follicle patterning

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The spatial patterning of mammalian hair follicle precursors in embryonic skin is most commonly studied in the laboratory mouse (*Mus musculus*), where new follicles form equidistantly from preexisting ones in successive waves. This simple geometric rule has been effectively described as emerging from an expansion-induction process. However, such a description is incompatible with more recent developmental data indicating instead that scale, feather, and hair placodes self-organize through reaction–diffusion–chemotaxis cell interactions involving epidermal and dermal signaling. Here, we suggest that the chemotactic component of this framework suffices to describe the dynamics of placode insertion in two mammalian species that exhibit drastically different patterns. More specifically, we investigate a continuum dynamical model capturing interactions between motile dermal mesenchymal cells and an epidermal chemoattractant, embedded in a two-dimensional, isotropically expanding domain representing the growing embryonic skin. Through numerical simulations, mathematical analysis, and comparison to experimental developmental data, we first show that the chemotaxis model gives rise to the effective geometric rule that initially justified the development of the expansion-induction model in the laboratory mouse. Second, we show that the strikingly regular hair placode pattern in the spiny mouse (*Acomys dimidiatus*)—with long-range order, specific orientation and anisotropies—is not generated by an expansion-induction mechanism, but is recapitulated by an anisotropic chemotaxis model combined with experimentally observed anisotropic growth. Overall, our findings reveal that variation in the chemotactic component of the corresponding self-organizational system might be a key determinant of interspecific differences in hair placode patterning dynamics and resulting spatial organizations.

chemotaxis | self-organization | hair follicle | patterning | development

With the exception of the scales on the face and jaws of crocodilians and on the top of the head in tortoises—that emerge from a mechanical compressive folding process (1, 2)—all vertebrate skin appendages, i.e., scales, feathers, and hairs, develop from anatomical skin structures named placodes. Placodes are typically characterized by a local epidermal thickening, an associated aggregation of underlying dermal cells, and conserved molecular signaling (3, 4). Hair and feather placodes—but not scale placodes—then further develop into a so-called follicular structure, characterized by the tubular invagination of the epidermis into the dermis (5, 6).

An important question pertaining to skin appendage development is how they are spatially patterned across the embryonic skin. In this respect, the spatio-temporal patterning of hair placodes/follicles in the embryonic skin of the laboratory mouse (*Mus musculus*) follows an intricate dynamical process (Fig. 1A). Indeed, the hair placode positions are established in three successive rounds—or so-called “waves”—of induction, giving rise to three hair types: the thickest and longest mechano-sensitive guard hairs developing around embryonic day 14 (E14), followed by the shorter awl hairs around E15; and then, after E16, the zigzag hairs characterized by multiple bends along their shaft (7, 8). During each wave, the newly induced placodes appear in positions equidistant from existing ones as the embryonic skin expands (Fig. 1A), such that the typical distance among neighboring follicles—i.e., the length scale of the pattern—is preserved during the process (8, 9).

Over the past decades, studies integrating experimental and theoretical approaches have shown that reaction–diffusion (RD) types of interactions among molecular activators and inhibitors are involved in the self-organized patterning of hair and feather placodes (11–14). More concretely, epidermal activators and inhibitors include Wntless-Integration (WNT) and Dickkopf (DKK) signaling proteins, respectively (12, 15–17), whereas dermal modulators include proteins of the Fibroblast Growth Factor (FGF) (18, 19) and Bone Morphogenetic Protein (BMP) families (11).

Significance

The dynamics of development and resulting spatial arrangement of hairs, feathers, and scales widely varies across vertebrate species. A classical description, known as expansion-induction, proposed that hair follicle precursors form in successive waves during embryonic skin growth because existing precursors inhibit the development of nearby ones. Our work instead identifies chemotaxis—the directed movement of cells toward chemical cues—as an alternative mechanism underlying this process. Through comparisons of computational simulations and experimental developmental data from two rodent species with contrasting follicle patterns, we show that chemotactic self-organization can generate, in isolation, a wide range of skin hair precursor arrangements. This framework underlines how simple cellular interactions can produce complex and diverse tissue architectures across species.

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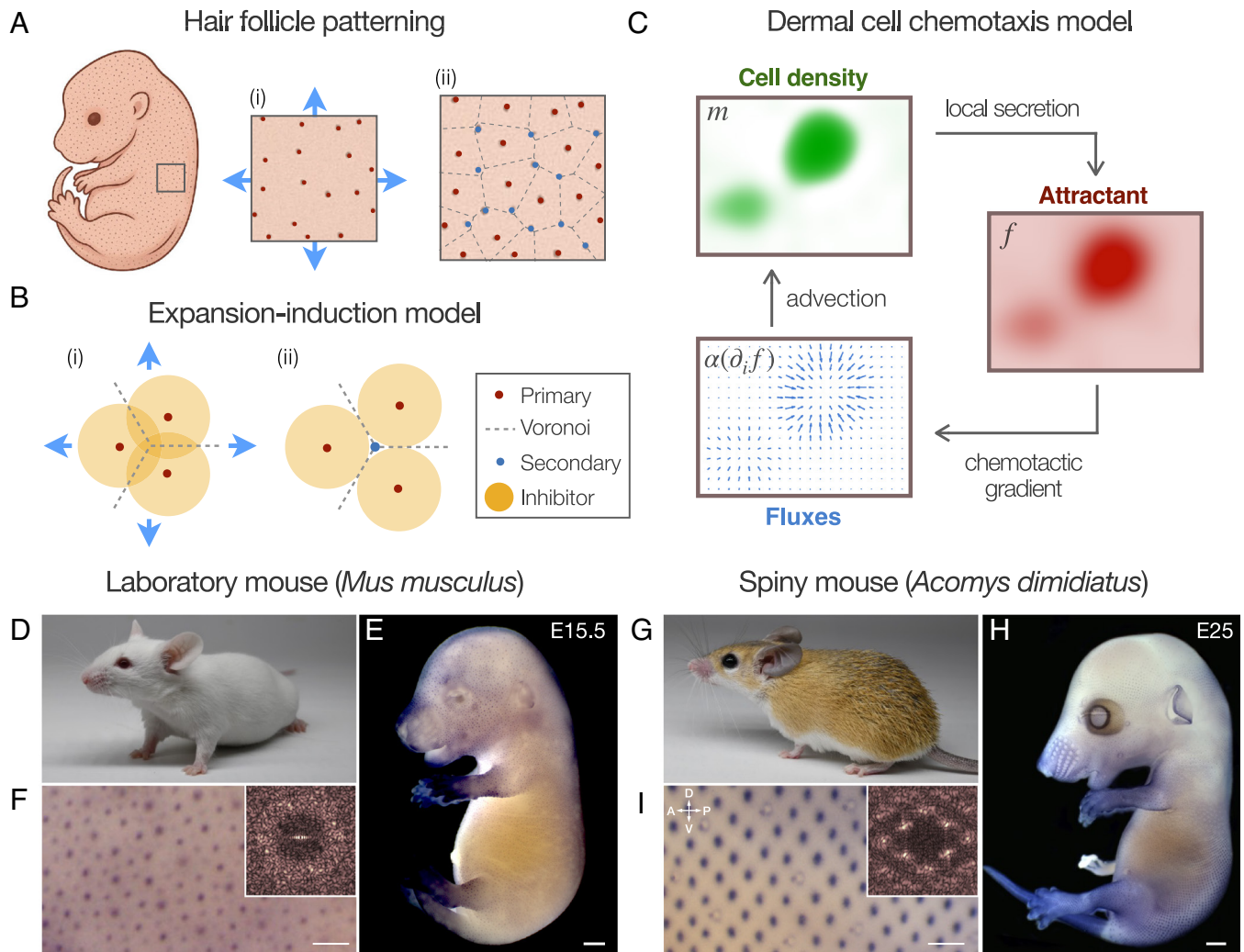


Fig. 1. Expansion-induction model (based on positional information) vs. the self-organizational chemotaxis model of placode patterning in two rodent species. (A) Schematics of the laboratory mouse hair follicle patterning dynamics: as the embryo grows (i), new follicles (blue dots) appear at equidistant positions (ii) between existing ones (red dots). (B) The expansion-induction model (10) is compatible with the geometric rule that characterizes mouse hair follicle patterning. (C) The chemotaxis model: a field density of motile mesenchymal cells, $m(\mathbf{r}, t)$, promotes the local secretion of a diffusive chemical attractant, $f(\mathbf{r}, t)$. The gradient of attractant gives rise to chemotactic fluxes which, in turn, advect the cell density toward high concentrations of the attractant. (D–F) Adult laboratory mouse and whole-mount in situ hybridization (*Ctnnb1*) in an E15.5 embryo. (G–I) Adult spiny mouse and whole-mount in situ hybridization (*Left1*) of an E25 embryo. [Scale bar, (E) and (H): 1 mm.] (F and I) show close-ups of the corresponding placode patterns (Scale bar, 250 μm). Insets in F and I show the Fast Fourier Transform (FFT) of the corresponding placode pattern.

Although theoretical RD models generate steady-state spot patterns with spatial statistics similar to those of the mouse placode pattern (12, 13), it has been proposed that another description, called the expansion-induction model (10), better predicts the sequential insertion of new hair placodes between existing ones as the embryonic skin expands (Fig. 1B). Indeed, given the consistent geometric pattern in which new follicles form at positions equidistant from existing ones (Fig. 1A), the expansion-induction model suggests that existing placodes act as cues of positional information that dictate where and when the new ones will be inserted (10). To account for such dynamics, each existing placode is assumed to be associated with an inhibitor source (Fig. 1B) that prevents nearby formation of new placodes and, importantly, does not scale with the system size. Hence, as the embryo grows and the interplacodal distances increase, skin regions with sufficiently lowered inhibitor density experience the induction of new placodes. This expansion-induction mechanism accurately predicts the position and timing of new placode insertions in the mouse, and hence, it captures the statistical properties of the resulting follicle

pattern (10). However, the suggested underlying positional information mechanism does not fit well with the known self-organized, spontaneous cellular and molecular interactions that govern the development of hair follicles.

Again, with the exception of crocodilian and tortoise head scales (1, 20)—self-organized periodic patterning of integumentary appendages in mammals, birds, and reptiles has been shown to involve not only RD types of molecular interactions (11, 12, 21, 22), but also the local chemotactic aggregation of dermal mesenchymal cells (21–27), which produces a positive feedback loop on placode induction through the compression of mechano-sensitive epidermal cells (28–30).

Here, we ask if the chemotactic component of such reaction–diffusion–chemotaxis models can, in isolation, explain the emergent insertion of units with spatio-temporal characteristics similar to those observed for mouse follicle patterning. In other words, we study whether chemotactic self-organization can give rise to the ad-hoc geometric rule on which the effective expansion-induction model is based. To this end, we investigate a simple continuum

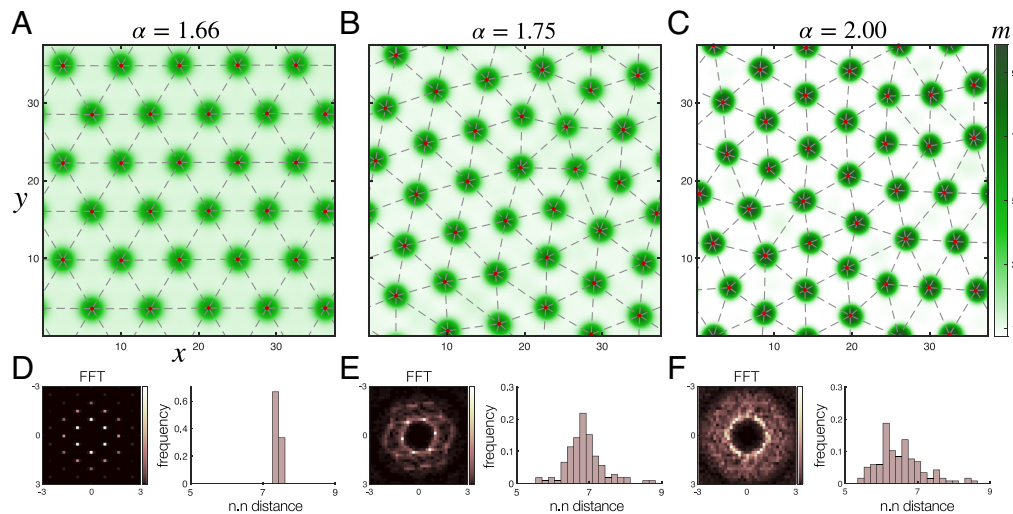


Fig. 2. Characterization of the asymptotic spot states in the chemotaxis model. (A) The regular spot pattern of a system with chemotaxis strength $\alpha = 1.66$ close to the onset of instability. The dark red dots denote the local maxima of m , indicating spot centers. Dashed gray lines denote the edges of the Delaunay triangulation among nearest neighbors. (B and C) Spot patterns of systems with (B) $\alpha = 1.75$ and (C) $\alpha = 2.00$. Other parameters are $D = 0.5$, $d = 0.5$, $\lambda = 0.25$. (D) Characterization of the spot pattern in A. *Left* panel shows the 2D FFT of the pattern, whereas the *Right* panel displays the distribution of nearest-neighbor distances. (E and F) Same analyses as in (D) for systems in panels (B and C), respectively.

model describing the interactions between dermal mesenchymal cells and a chemical attractant (Fig. 1C). First, we perform linear stability analysis (SI Appendix, section 2) to identify the parameter space where this model spontaneously yields spot patterns (Fig. 2), describing the spatial patterns of placodes. Second, we investigate the dynamics of new spot insertions in a domain undergoing pure isotropic expansion (Fig. 3). Using numerical simulations and mathematical arguments, we show that the chemotaxis dynamics recapitulate exactly those of the expansion-induction model in the mouse. Third, we test the accuracy of the chemotaxis model using embryonic mouse data, and find that this model predicts secondary/tertiary follicle positions with the same remarkable accuracy as the expansion-induction model (Fig. 4).

Next, using light-sheet imaging of spiny mouse embryos (*Acomys dimidiatus*), we investigate hair follicle patterning on the 3D surface of their skin (Figs. 1G–I and 5), where we find that the follicle positions are established through two successive waves of induction. In contrast to the laboratory mouse (Fig. 1D–F), the spiny mouse exhibits a strikingly regular and long-ranged pattern of placodes (Movie S1), whose induction dynamics do not follow the geometric rule of the expansion-induction model. Using numerical simulations, we show that the chemotaxis model can give rise to a spot lattice whose geometrical features—i.e., long-range order, as well as specific orientation and anisotropy relative to the animal body axes—are highly similar to those observed in the spiny mouse (Fig. 5). Remarkably, appropriate parametrization of the chemotactic model suffices to generate the correct order and orientation of the spiny mouse pattern, whereas anisotropic growth is also required to explain its anisotropic stretch along the anterior–posterior axis. Finally, we show that dermal cell nuclei and actin filaments are preferentially oriented along the dorso–ventral axis of the animal (Fig. 6), explaining the elongation of individual placodes because of increased diffusive displacements of mesenchymal cells along that same axis (Fig. 5). Taken together, our analyses indicate that, contrary to the expansion-induction model, the feedback loop generated by chemotactic self-organization alone is sufficient to capture salient features of hair follicle patterning in two species characterized by very different spatial orders.

Green and Sharpe have previously argued (32) that, contrary to common belief, the paradigms of positional information and self-organization are not mutually exclusive, but can combine their effects during morphogenesis. In addition to this perspective, we propose here that certain developmental processes traditionally hypothesized to occur according to the paradigm of positional information—e.g., the expansion–induction model of follicle patterning—may in fact arise from self-organized reaction–diffusion–chemotaxis dynamics, which can then be generalized to other species and contexts. Our work thus contributes to the growing recognition of the applicability of the reaction–diffusion–chemotaxis framework (25, 33, 34), while still acknowledging the relevance of positional information in other developmental scenarios.

Results

Continuum Model of Dermal Cell Chemotaxis. To describe the dynamics of hair follicle patterning in the skin through chemotactic self-organization, we investigate the interactions between two continuum scalar fields (Fig. 1C): a concentration density of motile mesenchymal cells in the dermis, $m(\mathbf{r}, t)$, and a chemical attractant density (e.g., the Fibroblast Growth Factor, FGF), $f(\mathbf{r}, t)$. The latter is produced by epidermal cells and promotes the aggregation of mesenchymal cells, but for simplicity, the dermis and epidermis are collapsed into a 2D surface.

The mesenchymal cell dynamics is described by the following equation:

$$\frac{dm}{dt} = D_m \partial_i^2 m - d_m m + p_m - \alpha \partial_i (m e^{-\lambda m} \partial_i f). \quad [1]$$

Throughout, we use ∂_i to denote partial derivatives with respect to the spatial coordinate $r_i \in \{x, y\}$ and adopt the Einstein notation of summing over repeated indices. The left-hand side of Eq. 1 denotes the material time derivative of m , whereas D_m , d_m , and p_m on the right-hand side denote the diffusion, degradation, and production constants of m , respectively. While D_m characterizes the amplitude of random cell motility, d_m and p_m describe the cell turnover kinetics, i.e., the rates of random cell death and cell production, respectively. Last, the fourth right-hand side term in Eq. 1

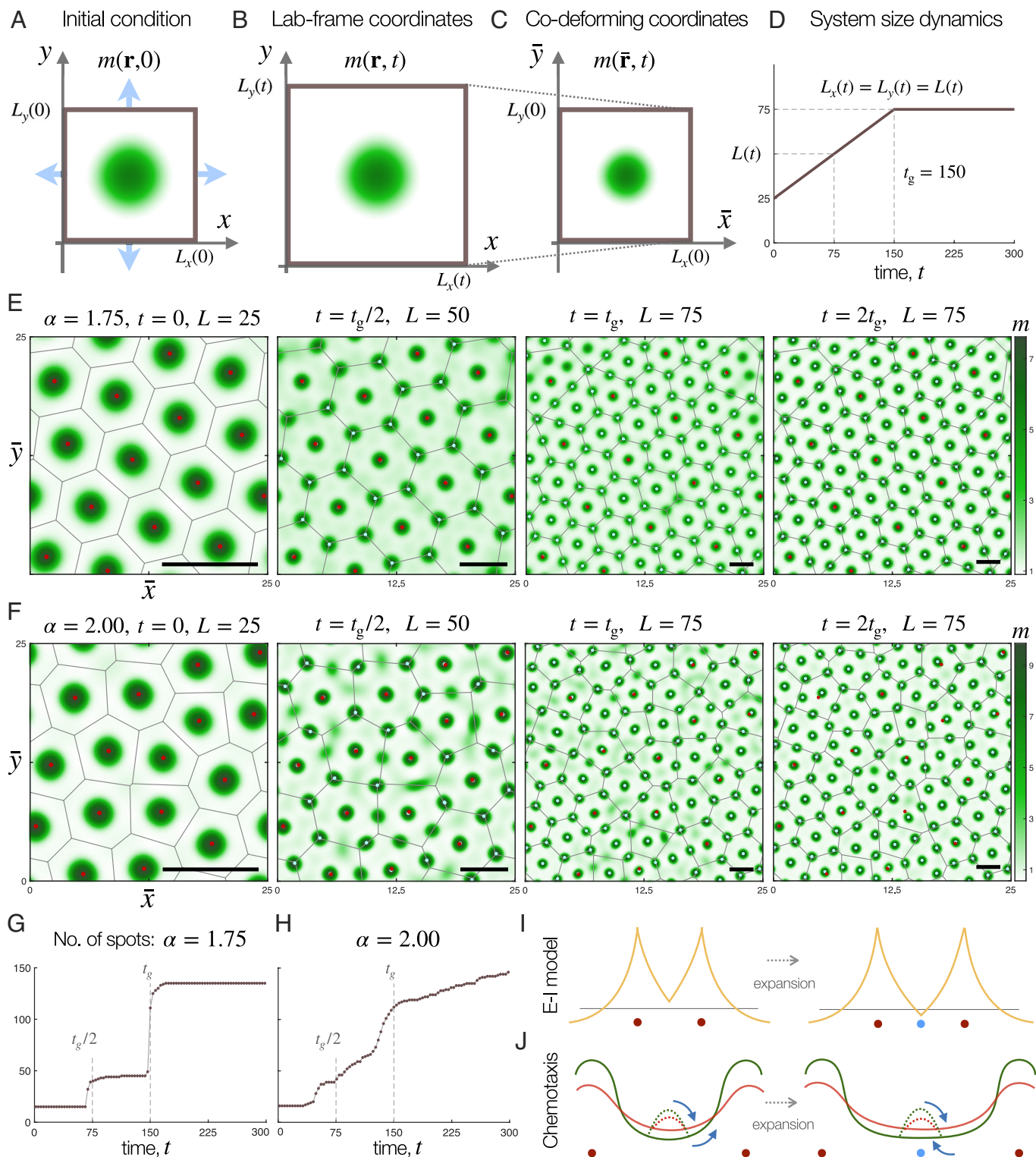


Fig. 3. Spot insertion dynamics in the chemotaxis model under isotropic expansion. (A–C) We perform numerical simulations in the co-deforming coordinates. A system of initial length L_0 (A) undergoes isotropic expansion in time. At every time point, we map the lab-frame coordinates (B) to the co-deforming coordinates (C). (D) L dynamics for a square system (SI Appendix, section 1.2). (E) Dynamical trajectory of a quasi-hexagonal spot pattern in an expanding system with $\alpha = 1.75$, displayed at different times. Gray solid lines denote the Voronoi boundaries of spots positions at $t = 0$ (Left panels) and at $t = t_g/2$ (Right panels). Red dots denote spot centers at $t = 0$, cyan dots are the current spot centers. A new spot is inserted when the corresponding local maximum reaches 0.9 times the maximal value of m . Black scale bars correspond to the same length. (F) Trajectory as in (E) but for a less regular pattern, where $\alpha = 2.00$. (G and H) Number of spots over time for systems with $\alpha = 1.75$ (G) and $\alpha = 2.00$ (H). (I and J) Schematic representation of the spot insertion dynamics in the expansion-induction model (I) and the chemotactic model (J) (blue arrows indicate cell density fluxes).

describes the effect of chemotaxis (24, 25), where directed motility of cells is induced toward ($\alpha > 0$) or against ($\alpha < 0$) the gradient of the chemical attractant, $\partial_x f$. In this study we focus on systems

with $\alpha > 0$, for which there is experimental evidence (23). Note that the exponential $e^{-\lambda m}$ describes a nonlinear “jamming factor,” i.e., that the cell density has an accumulation limit controlled by λ .

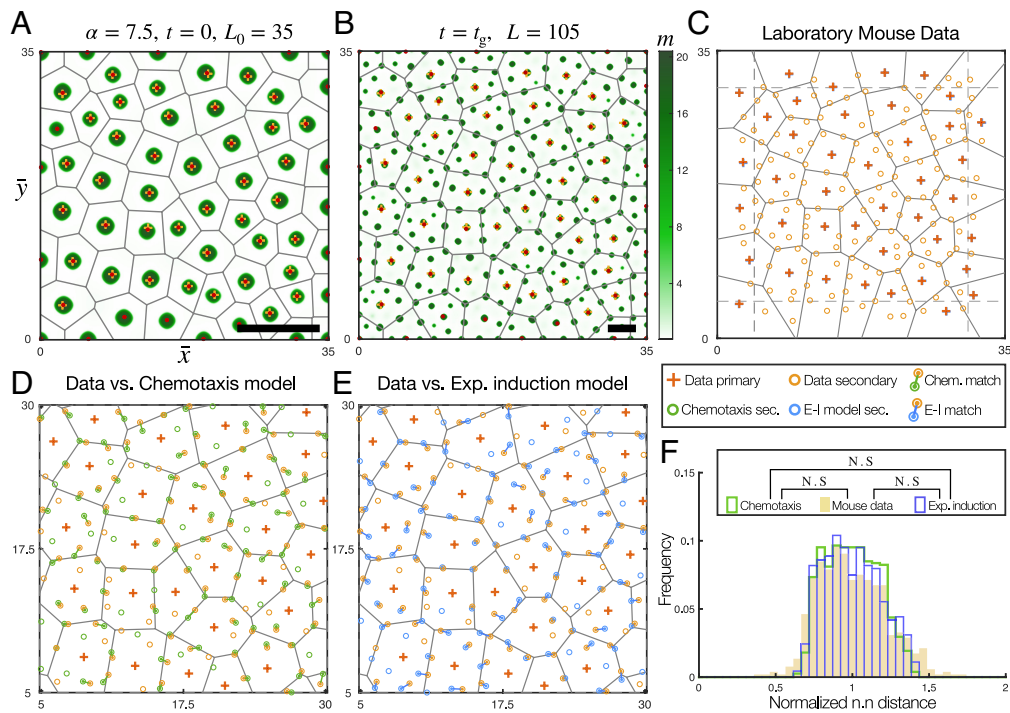


Fig. 4. Predicting the observed embryonic laboratory mouse follicle positions with the chemotaxis and expansion-induction models. (A) A stationary state of the chemotaxis model with fixed sources (orange crosses) achieves the same pattern as primary follicles in experimental data. (B) System state after expansion, with $t_g = 150$. (C) Observed embryonic mouse data with primary and secondary/tertiary placodes displayed as orange crosses and circles, respectively. We focus on the region inside the gray dashed square to avoid boundary artifacts. (D and E) Matching of the embryonic data with the chemotaxis (D) and expansion-induction (E) (from ref. 10) model predictions. (F) Distributions of the normalized nearest-neighbor distances for the mouse data (light orange area), the chemotaxis model (green bars), and the expansion induction model (blue bars). The two-sample Kolmogorov–Smirnov (K-S) test reveals no statistically significant difference between the three distributions (SI Appendix, section 10).

On the other hand, the dynamics of the chemical attractant f follows a classical RD equation:

$$\frac{df}{dt} = D_f \partial_i^2 f - d_f f + p_f m, \quad [2]$$

where, D_f , d_f , and p_f denote the diffusion coefficient, the degradation rate, and the secretion rate of f , respectively. Note that epidermal, rather than mesenchymal cells, secrete f . However, because this secretion rate depends [through mechanical feedback (28)] on the amount of aggregated mesenchymal cells, we set the rate p_f proportional to m . Eqs. 1 and 2 are based on the well-known Keller–Segel model (35), and similar forms have already been used to describe hair, feather, and scale primordia patterning in vertebrate embryos (22, 24, 25, 36–38). In contrast to most previous works, we focus here on the dynamics of this model in an expanding domain.

After rescaling space, cell and attractant densities by $\tilde{r}_i = \sqrt{d_f/D_f} r_i$, $\tilde{f} = d_f^2/(p_f p_m) f$, $\tilde{m} = d_f/p_m m$, and $t = d_f t$ respectively, we get the following set of nondimensional equations:

$$\frac{dm}{dt} = D \partial_i^2 m - dm + 1 - \alpha \partial_i (m e^{-\lambda m} \partial_i f), \quad [3]$$

$$\frac{df}{dt} = \partial_i^2 f - f + m. \quad [4]$$

Here we have dropped the ‘tilde’ sign in every term and end up with four effective parameters: i) the diffusion and ii) degradation ratios, $D = D_m/D_f$ and $d = d_m/d_f$; as well as the accordingly rescaled iii) chemotactic strength α , and iv) jamming factor λ (SI Appendix, section 1). First, we study asymptotic states of the

nondimensionalized system in a square domain ($L_y = L_x = L$) with fixed size and periodic boundary conditions, unless stated otherwise. Next, we investigate the transient dynamics of this model in a domain undergoing pure isotropic expansion. Detailed description of nondimensionalization and a full specification of the model is provided in (SI Appendix, section 1).

Spots as Asymptotic States of the Chemotaxis Model. We start with linear stability analysis to identify the parameter space where the chemotaxis model yields spontaneous patterns. To this end, we study how small perturbations affect the homogeneous steady state (h.s.s.) (SI Appendix, section 2), where we compute the growth rate of each perturbation mode depending on its wavelength and the parameters of the system (SI Appendix, Fig. S1). Such stability analysis reveals that, regardless of other parameters, the chemotaxis model yields spontaneous patterns for sufficiently large values of the chemotaxis strength α .

Next, we perform numerical simulations of the system for different values of α , while keeping D , d and λ parameters fixed, and find that the system can give rise to spots as stable nonlinear states. In particular, we investigate how the asymptotic spot states patterns change as the system is driven away from the onset of instability. Initially, we find that the spots form a nearly regular hexagonal lattice when $\alpha = 1.66$ (Fig. 2A), i.e., when the chemotaxis strength is just above the critical value required for spontaneous patterning (SI Appendix, Fig. S1A). We note that the spacing distance between nearest neighbors (Fig. 2A) is consistent with the patterning length scale derived from the linear stability analysis (SI Appendix, section 2.3). Second, we observe that the spot patterns become increasingly disordered as α is increased (Fig. 2 B and C). This is due to the fact that a higher α allows for

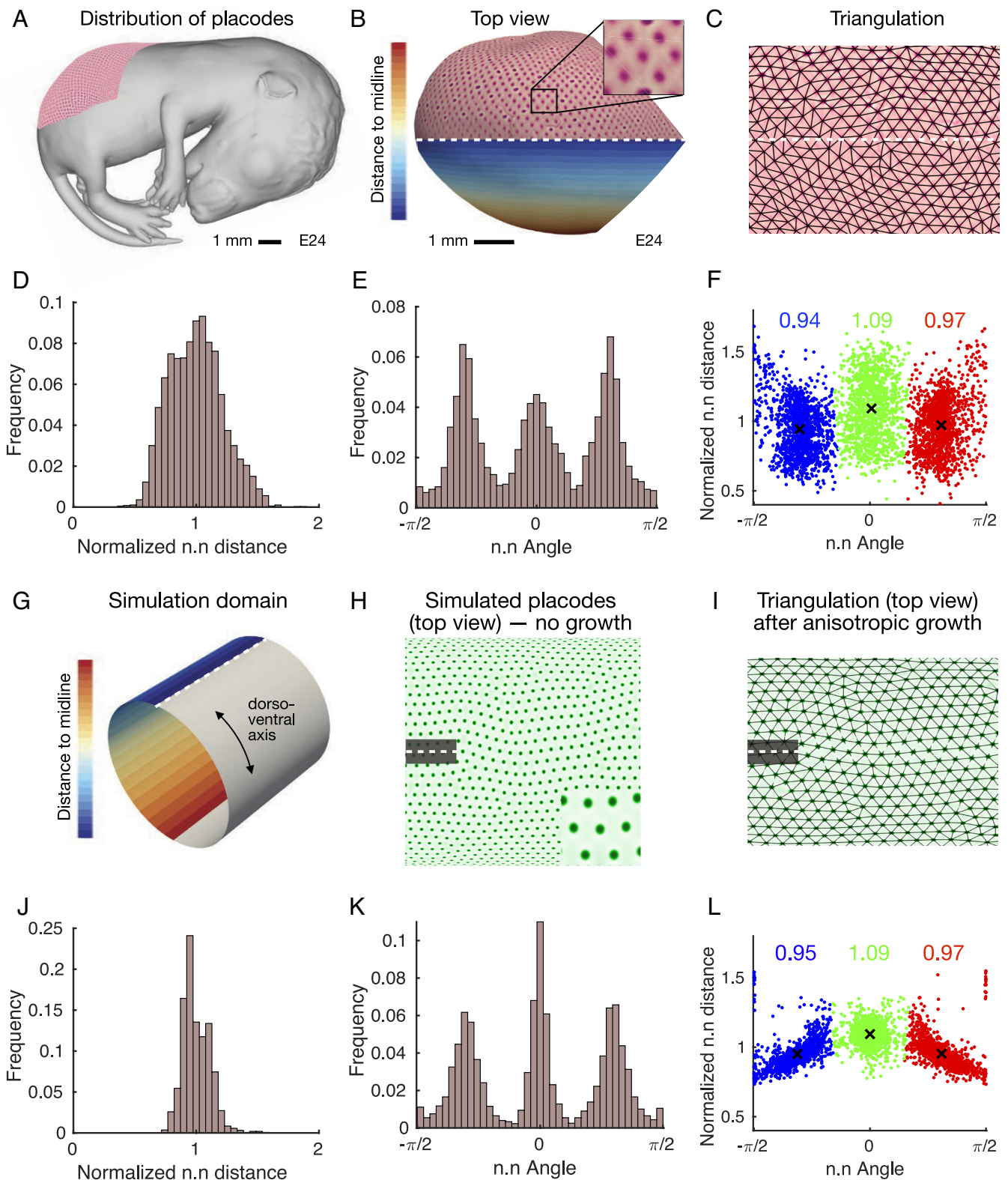


Fig. 5. The anisotropic chemotaxis model captures follicle patterning dynamics in the spiny mouse. (A and B) Positions of primary and secondary placodes are identified on the lower-back 3D surface of an E24 embryo (imaged with light-sheet microscopy; [Movie S1](#)) using nuclear staining. The color map in (B) shows the geodesic distance from the midline (dashed white line), and the *Inset* shows that placodes are slightly elongated along the dorso-ventral axis. (C) Placode lattice shown on the 3D surface of the embryo, obtained via the Ball-Pivoting algorithm (31). (D and E) Length distribution (D) and angle distribution (E) of triangulated edges formed by the placode positions. (F) Dependence of the edge length on edge angle. Colored numbers and black crosses indicate mean nearest-neighbor distances and the corresponding centroids computed using K-means clustering. (G) Cylindrical model geometry with geodesic distance from the midline. (H) Simulated placode positions. (I) Triangulation of the placode lattice using the Ball-Pivoting algorithm, after anisotropic growth (factor of 1.46 along the cylinder axis; [SI Appendix, Fig. S10](#)). Chemotaxis model parameters ([SI Appendix, section 4.2](#)): $\alpha_{\parallel} = 2.46$, $\alpha_{\perp} = 1.2\alpha_{\parallel}$, $d = 1$, $\lambda = 0.35$, $D_{\parallel} = 0.1$, $D_{\perp} = 1.5D_{\parallel}$, where $\parallel$ denotes the cylinder axis. (J–L) Edges of spot centers in (I), displayed as in (H). (J–L) Statistical analyses of simulated patterns, as computed in panels (D and E).

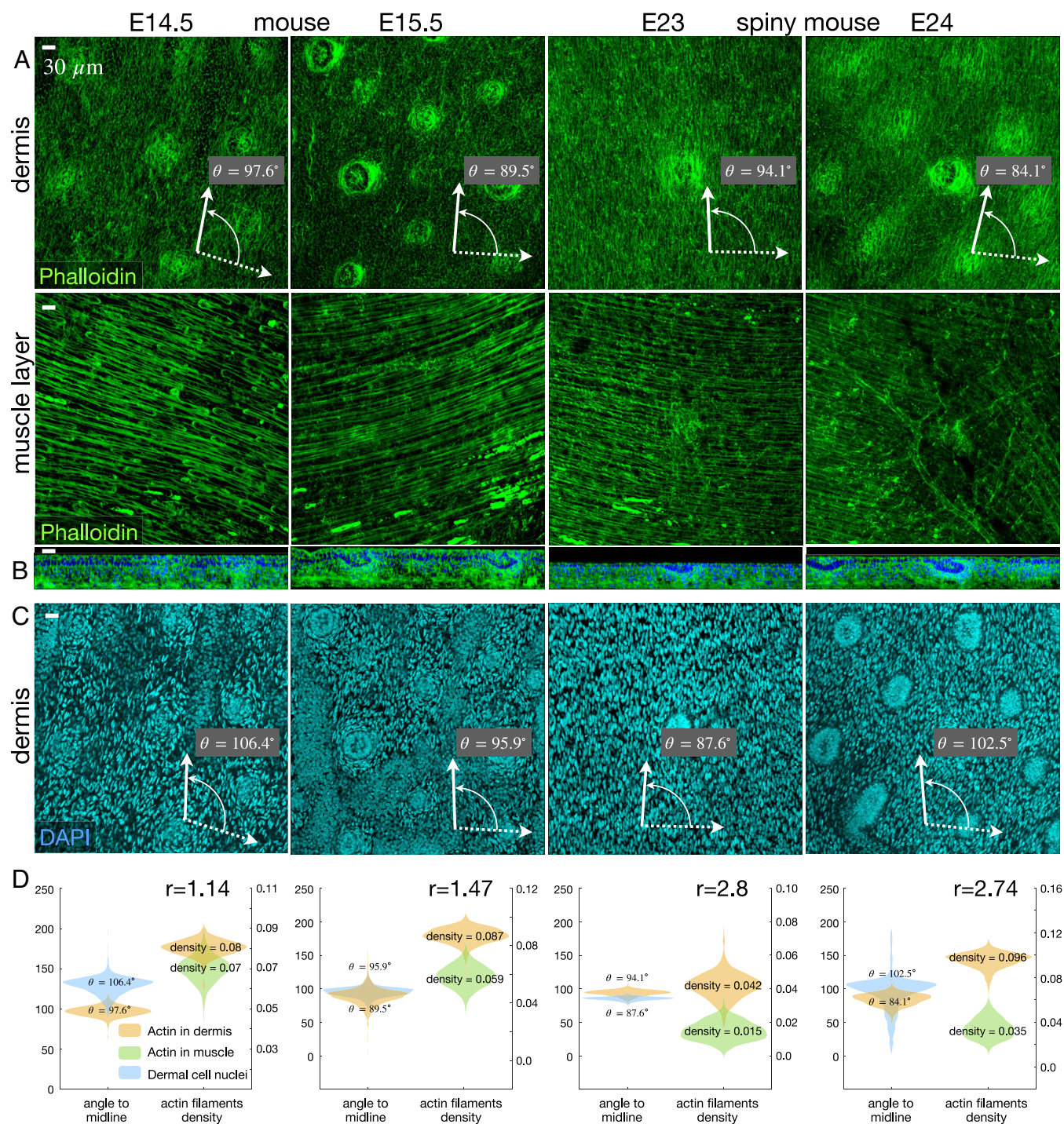


Fig. 6. The orientation of actin filaments in the spiny mouse embryonic dermis supports anisotropic displacement of mesenchymal cells. (A) Phalloidin staining (green) of dermal actin filaments in the laboratory mouse and the spiny mouse embryonic skin shown as maximum projection intensity. The mean orientation of actin fibers (plain arrow) in mesenchymal cells is approximately at 90° with respect to the midline (dashed arrow), i.e., the actin fibers have a preferential dorso-ventral orientation. The lower panels show the actin fibers in the underlying muscle layer. (B) Cross section of the same phalloidin-stained samples with DAPI (blue) staining nuclei. (C) The nuclei of mesenchymal cells are elliptic and the mean orientation of their main axis is similar to that of actin fibers. (D) Distribution of dermal actin filaments and nuclei orientations (left column of each graph) and densities of dermal and muscle actin filaments (right column of each graph; the ratio is also shown) in the embryonic skin of the mouse and the spiny mouse at two developmental stages. [Scale bar, (A–C): $30\ \mu\text{m}$.]

a larger domain of perturbation modes to affect the late-time patterning.

To characterize the regularity of spots in the nonlinear states, we initially use two measures. First, we compute for each pattern its corresponding 2D Fast Fourier Transform (FFT; left maps, Fig. 2 D–F). If the spot pattern is regular, the Fourier transform corresponds to the Bravais lattice of the pattern (39). This is the case for $\alpha = 1.66$, where the Fourier Transform consists of sharp

peaks forming a hexagonal pattern (Fig. 2 D, Left panel). Conversely, the loss of pattern regularity as α increases is reflected by fuzzier Fourier peaks (Fig. 2 E and F, Left panels). Second, we compute for each pattern the distribution of nearest-neighbor distances in the Delaunay triangulation among spot centers. Note that the center of each spot (red dots, Fig. 2 A–C) is defined as the local maximum density of m . As expected, the distribution of neighbor distances is very localized around the mean value for the

regular pattern, and becomes increasingly more dispersed in systems with higher chemotactic strength α (Fig. 2 D–F, Right panels).

Overall, these analyses indicate that chemotactic self-organization gives rise to nonlinear states of spots that exhibit a variety of spatial arrangements: from patterns characterized by low order and a typical minimal distance (as in the laboratory mouse; Fig. 1F) to patterns displaying long range order (as the lattice of placodes in the spiny mouse; Fig. 1I and Movie S1).

Chemotaxis Model Simulations in an Expanding Domain. To investigate the dynamics of new spot formations, we perform numerical simulations of the chemotaxis model in a system under isotropic expansion. To this end, we adapt an approach from previous studies (40, 41) to define the “co-deforming” coordinates (SI Appendix, section 3), $\bar{\mathbf{r}} = (\bar{x}, \bar{y})$, i.e., a coordinate system that maps the system to the laboratory-frame coordinates, $\mathbf{r} = (x, y)$, in the following way:

$$\bar{\mathbf{r}}_i = \frac{L_0}{L(t)} \mathbf{r}_i,$$

where $L(t)$ denotes the time-dependent length of the system, and $L_0 = L(t = 0)$. Thus, while an expanding system in lab-frame coordinates (Fig. 3 A and B) is defined in $\mathbf{r}_i \in [0, L(t))$, in co-deforming coordinates (Fig. 3C) it is mapped to $\bar{\mathbf{r}}_i \in [0, L_0)$. In other words, as the system expands, the co-deforming coordinates continuously rescale the lab-frame coordinates, $\mathbf{r} = (x, y)$, to the initial size of the system. A full derivation of Eqs. 3 and 4 in co-deforming coordinates is detailed in (SI Appendix, section 3).

Using the chemotaxis model in co-deforming coordinates, we simulate dynamic trajectories of spot emergence as the system size is increased. First, we run simulations of the perturbed h.s.s. in a system of fixed size ($L_0 = 25$) until the system reaches an asymptotic state of spots (see SI Appendix, section 4.1 for numerical methods). We then allow $L(t)$ to increase from L_0 to a final length $L_f = 3L_0$ in a time $t_g = 150$ (Fig. 3D). Note that t_g is selected so that the time scale of expansion is slower than the time scale of pattern formation (SI Appendix, section 4.1), and the relaxation time is ensured to be long enough for the new stationary state to be reached. Note that we also display the dynamics of the chemotaxis model simulations in laboratory-frame coordinates in SI Appendix, Fig. S3.

More concretely, we investigate the dynamics of new spot formation at two points of the parameter space: on a regular pattern emerging from a chemotactic strength $\alpha = 1.75$ (Fig. 3E), and a less regular pattern that emerges from a system with $\alpha = 2$ (Fig. 3F), as introduced in Fig. 2. As the system expands, we observe that some interspot regions acquire higher cell density compared to the background levels. In addition, we observe that such regions correspond to the edges/vertices of the Voronoi diagrams of primary spots and, by the time that $L(t) \approx 2L_0$, these regions form the second generation of spots inserted in the system (cyan dots in second panels of Fig. 3 E and F). As the system size approaches $L(t) \approx 3L_0$, we observe the emergence of the third generation of spots which, again, appear at the edges/vertices of the Voronoi diagrams that include primary and secondary spots (third panels of Fig. 3 E and F). These analyses indicate that new spots appear equidistantly from existing nearest-neighbor spots. In the case of the more regular spot pattern (Fig. 3E), the new spots form at the vertices of Voronoi diagram formed by previously existing units, reflecting the fact that every new spot is inserted at equal distance from three existing mutually first-neighbors. In the less regular spot pattern (Fig. 3F), the position of emerging new spots is less constrained by the configuration of existing spots:

new spots form not only at the vertices but also at the edges of the corresponding Voronoi diagram. Overall, our analyses provide the first evidence that the geometric rule observed in the laboratory mouse—which justified the expansion-induction model (10)—might actually emerge from the self-organizing chemotactic component of a reaction–diffusion–chemotaxis system under isotropic expansion.

Moreover, we investigate the relative timing of new spot insertions in the chemotactic system with different pattern regularities. In the more regular pattern ($\alpha = 1.75$), we find that new spots appear almost synchronously, i.e., within short time intervals (Fig. 3G). This is because interstitial spaces between existing spots are nearly equivalent to each other, such that new spots of high densities form in equivalent positions at about the same time. The timing of spots becomes more asynchronous in the less regular pattern ($\alpha = 2$, Fig. 3H), where the new spots form around the Voronoi vertices before forming around the Voronoi edges (e.g., Fig. 3F, second panel). These observations indicate that the geometry of the primary spots, not only determines the positions of new spots, but also the timing of their emergence. Intuitively, this occurs because, in an expanding system, primary neighboring spots with larger-than-average neighbor distances will “accommodate” a new spot earlier than their counterparts with smaller neighbor distances. Hence, a more irregular primary pattern gives rise to a “smoother” temporal slope (Fig. 3H).

Placode Insertion Dynamics Emerge from Chemotactic Self-Organization. Our results suggest that the insertion dynamics predicted by the expansion-induction model (10) can instead be explained by the underlying chemotaxis system. The expansion-induction model assumes that the expansion of the domain causes the distances between neighboring inhibitor sources (i.e., the existing placodes) to increase, such that new spots are inserted in regions where the inhibitor density falls below a threshold. If we consider a simple 1D scenario of two identical sources (red dots, Fig. 3I), the resulting inhibitor concentration has a minimum exactly between the existing source points. As the system expands, this minimum will be the first point to drop below the concentration threshold, and the new point source will be inserted exactly between the existing ones (blue dot, Fig. 3I).

To understand how these dynamics may emerge from chemotactic self-organization, we study analytically a peak-forming one-dimensional (1D) chemotaxis system that continually expands by infinitesimal amounts (Fig. 3J). Our mathematical analysis (SI Appendix, section 5) reveals that, as the system expands, the new peak indeed starts forming at a point with previously minimal values of m and f concentrations, corresponding to the mutually equidistant point from existing peaks (Fig. 3J and SI Appendix, Fig. S4). Importantly, we show that this phenomenon is insensitive to the model parameter values, provided that the system is in a pattern-forming region. In other words, the 1D chemotaxis peak patterns have a generic response to slow expansions of the system, and this response is equivalent to that of the expansion-induction model. Moreover, our additional simulations of the chemotaxis model in 2D confirm the analytical prediction that such spot insertion dynamics is indeed robust to parameter variations (SI Appendix, Fig. S5).

Note that one can develop an intuitive understanding of how the chemotaxis term (Eq. 1) drives the insertion dynamics of new spots equidistantly from existing ones. Indeed, the local f gradients attract diffusive cells toward existing spots and, as a consequence, the interstitial regions between spots have minimal cell density and corresponding minimal attractant values. Consequently, the magnitude of the f gradient in these regions is smaller (Fig. 3J, Left panel) and

the chemotactic fluxes are lower relative to other regions. As the system expands, chemotaxis in the interstitial region decreases even further, such that new dermal cells m produced by the system do not migrate but instead accumulate, forming a new local maximum of cell density (Fig. 3 J, Right panel). This new local m maximum leads to the production of a colocalized f maximum attracting surrounding cells. Beyond this point, the positive feedback loop between m and f will establish a new interstitial spot of equal cell/attractant density with the already existing spots. The mathematical argument behind this intuition is provided in (SI Appendix, section 5).

Validation of the Chemotaxis Model with Laboratory Mouse

Data. In this section, we use available experimental data (from ref. 10) to evaluate the power of the chemotaxis model in predicting the positions of secondary and tertiary placodes (Fig. 4). In particular, we follow the same simulation procedure as in Fig. 3, but introduce the following modifications to appropriately integrate the laboratory mouse data. First, we introduce the positions of experimentally observed primary placodes as fixed localized sources of m (SI Appendix, section 6). Then, we allow the system (held at a fixed size) to reach its nonlinear spot state, and confirm that the spot centers (red dots in Fig. 4 A and B) coincide with the prescribed sources (orange crosses), corresponding to the primary placodes. Here, we impose no-flux boundary conditions (SI Appendix, section 1.1 and section 2.4) and use the linear stability analysis to select a parameter set that supports the interplacode spacing of the experimental data. The localized sources are maintained throughout the simulation to reflect the relatively advanced morphogenetic state of primary follicle precursors when secondary placodes form—i.e., at this stage, primary placodes have fixed relative positions because they have transitioned to hair pegs.

Next, we run simulations where we increase L from $L_0 = 35$ to $L_f = 3L_0$ [i.e., the observed skin growth at the corresponding developmental stages (42)] and record the positions where new spots appear during the simulation (Fig. 4B). Note that, as in the expansion-induction model case, we allow the chemotaxis system to evolve until the number of newly formed spots is approximately equal to that observed in the experimental data. We then compare the positions of secondary and tertiary placodes observed in the laboratory mouse (Fig. 4C) with those of placodes simulated with the chemotaxis model (Fig. 4D) and with the expansion-induction model (Fig. 4E). To this end, we use the following procedure: focusing on a central region of the domain (dashed gray lines in Fig. 4C) to mitigate boundary artifacts, we match observed (orange circles) and simulated points one to one on the basis of two criteria: i) their distance must be below a defined threshold (SI Appendix, Fig. S6), and ii) they must be mutual nearest neighbors. The results of this matching procedure are shown in Fig. 4 D and E for the chemotaxis and expansion-induction models, respectively.

Overall, we find that the chemotaxis model predicts the positions of secondary and tertiary follicles of the embryonic laboratory mouse with the same precision as the expansion-induction model. Indeed, in both cases, the proportions of simulated points that match observed secondary/tertiary follicles inside the region of interest are $\approx 80\%$ (86/110). Moreover, the nearest-neighbor distance distributions of the data observed in the mouse as well as simulated with the chemotaxis and expansion-induction models are statistically indistinguishable (Fig. 4F), indicating that the three patterns share the same variation in spacing. Note that the distribution of experimental mouse data (orange bars in Fig. 4F) appears slightly wider compared to both chemotaxis and expansion-induction distributions (green and blue bars). This difference may be due to multiple factors, including the projection of the curved embryonic skin onto

the 2D images used for analysis, as well as potential heterogeneity and/or anisotropy in skin expansion, or temporal asynchronicity in follicle induction (43) across the investigated domain. In addition, both chemotaxis and expansion-induction models are defined in 2D, and therefore neglect vertical cell movements (44), out-of-plane signaling (23), epithelial–mesenchymal cross-talk (45), or volumetric growth.

Finally, we note that, in both chemotaxis and expansion-induction models, the secondary spots match, on average, the real data better than tertiary spots (SI Appendix, Fig. S6), indicating error propagation (10). To investigate this further, we perform an additional simulation of the chemotaxis model in which, contrary to the previous case, we introduce localized sources not only at the primary placode positions, but also at each newly forming spot as the system expands (SI Appendix, Fig. S7). While this additional positional constraint slightly increases the number of matched follicles (89/110 instead of 86/110), the secondary spots still match better than the tertiary ones (SI Appendix, Fig. S7). Taken together, these results demonstrate that the chemotaxis and expansion-induction models perform comparably in reproducing the spatial patterning of laboratory mouse hair placodes.

The Spiny Mouse Hair Pattern Validates the Chemotaxis Model.

Spiny mice are characterized by stiff, spine-like hairs on the back (46) with a strikingly long-range ordered spatial pattern (Fig. 1I). While there has been increasing interest in spiny mice in the context of regeneration, diabetes, obesity, and reproductive physiology (47, 48), little is known about their embryonic development, in particular, how their highly ordered lattice of hair follicles is generated. To investigate the dynamics of placode patterning in spiny mice, we performed whole-mount in situ hybridizations with species-specific probes targeting the lymphoid enhancer-binding factor 1 (*Left1*) transcript. This transcription factor, involved in the WNT canonical signaling pathway, is essential in promoting placode and follicle formation and, thereby, constitutes an excellent marker of early hair morphogenesis (5, 49–51). At embryonic days 22 and 23, the first wave of follicle induction spreads from the lower back in all directions (SI Appendix, Fig. S8 A and B). The second wave then starts on the lower back, with faint *Left1* expression first visible as rings—each surrounding a primary placode—that in certain cases appear as rhombuses (SI Appendix, Fig. S8C). Eventually, expression concentrates at the nodes and edge midpoints of each rhombus, generally forming six to eight new placodes (Fig. 1I and SI Appendix, Fig. S8 E and F). As a result, and starting at E24, the densely distributed secondary placodes arrange at regular intervals along lines oriented diagonally with respect to the dorsoventral axis.

Here, we investigate how effectively the expansion-induction and the chemotaxis models capture these dynamics leading to a highly regular grid of placodes. First, we find that the expansion-induction mechanism does not explain the patterning of secondary placodes in spiny mice. We initially identify, on the lower-back of an E25 embryo (SI Appendix, Fig. S8 D–F), that the primary and secondary follicle precursors can be distinguished from their levels of *Left1* expression—low and high, respectively. We then plot the Voronoi diagram defined by primary placodes (crosses) and observe that the secondary placodes (circles) do not lie close to Voronoi nodes/edges (SI Appendix, Fig. S8G). Instead, despite a rather irregular pattern of primary follicles, the secondary placodes form a much more ordered grid than predicted by the Voronoi diagram, indicating that the patterning in spiny mice does not depend on an expansion-induction mechanism. This conclusion holds even when introducing anisotropic diffusion of the inhibitor in the expansion-induction model.

Second, we characterize the spatial statistical properties of the spiny mouse long-range ordered pattern on the curved 3D surfaces extracted from light-sheet imaging of an E24 embryo (Fig. 5A–C and Movie S1). Contrary to the laboratory mouse (SI Appendix, Fig. S9), the spiny mouse placode lattice exhibits a narrow normalized nearest-neighbor distance (Fig. 5D) and a trimodal distribution of the angles that nearest neighbors form (Fig. 5E) with respect to the anterior–posterior axis. These two statistics reflect the long-range hexagonal ordering of the spiny mouse pattern. In addition, this hexagonal pattern is “squeezed” as the edge lengths along the anterior–posterior axis are systematically larger (Fig. 5F). In other words, the spacing between nearest neighbors along the anterior–posterior direction of the embryo is larger than in the two other symmetry-equivalent directions of the hexagonal lattice. Finally, we observe that the spiny mouse placodes take an elliptical shape with the ellipses’ long axis being oriented along the dorso–ventral axis of the body (Fig. 1I and inset of Fig. 5B). Conversely, placodes in the laboratory mouse are circular (Fig. 1F and inset of SI Appendix, Fig. S9B). Overall, these analyses indicate that the lattice of placodes in the spiny mouse is characterized by a long-range order and elongated spacing along the anterior–posterior axis of the body, whereas the placodes themselves exhibit a shape anisotropy oriented dorso–ventrally.

Whereas these salient geometrical features are incompatible with the expansion-induction model, we investigate whether they can be captured by the chemotaxis model. We suggest that these features may emerge due to: *i*) anisotropic growth/deformation of the skin after patterning; and/or *ii*) anisotropic motility/chemotaxis of dermal mesenchymal cells. First, we analyze 3D light-sheet microscopy images of laboratory mouse and spiny mouse embryos at the two stages (E14.5 and E15.5 in the laboratory mouse; E23 and E24 in the spiny mouse) that correspond to the onset and the completion of secondary placode development, respectively (SI Appendix, Fig. S10). These analyses indicate that the embryos of both species grow more in the anterior–posterior than in the dorso–ventral axis. The ratio of growth along these two body axes is substantially larger (1.46) in the spiny mouse than in the laboratory mouse (1.16). Second, we perform simulations implementing anisotropies in the diffusion D and the chemotactic strength α (see SI Appendix, section 4.2 for derivation and numerical methods).

When realistic anisotropic growth as well as anisotropies in both the diffusion D and the chemotactic strength α (both increased along the dorso–ventral axis) are introduced, our numerical simulations on 3D surfaces (Fig. 5G–I) generate a lattice of spots whose features are highly similar to those of the spiny mouse (Fig. 5J–L). We note that the realistic shape of the spots (i.e., elongated along the dorso–ventral axis; inset of Fig. 5H) is due to the increased D along that axis, while the overall small values of α explains the long-range order (Fig. 2). Finally, the anisotropic lattice elongation along the anterior–posterior axis (Fig. 5L) is caused by the combination of anisotropic α (larger along the dorso–ventral axis) and anisotropic growth (greater in the anterior–posterior axis). We note that introducing anisotropic D alone can produce spot patterns with lattice orientation and spacing statistics similar to those observed in the spiny mouse, but with incorrect orientation of the elliptic spots (SI Appendix, Fig. S11). In contrast, anisotropic α alone (increased in the dorso–ventral axis) yields patterns with correct orientation of the lattice but both incorrect spacing statistics and incorrect orientation of the elliptic spots (SI Appendix, Fig. S12). Overall, these observations indicate that a combination of anisotropic diffusion (i.e., anisotropy in random cell motility) and anisotropic chemotactic strength is required for a proper description of the spiny mouse follicle patterning.

Remarkably, increased diffusion D and increased chemotactic strength α (both along the dorso–ventral axis) are compatible with the observed preferred orientation of the mesenchymal cells—i.e., the cell types subjected to chemotaxis. Indeed, we observe a thick and dense layer of dorso–ventrally oriented actin filaments in the upper dermis of the spiny mouse skin (Fig. 6A and B). Furthermore, we observe that dermal cell nuclei exhibit a similar preferential orientation (Fig. 6C). These two observations support the possibility that both the diffusive displacement of dermal cells (corresponding to the parameter D) and the chemotactic parameter α are increased along the dorso–ventral axis. Although oriented actin filaments are also present in the skin of the laboratory mouse at equivalent stages of placode formation, their density in the spiny mouse embryonic dermis is much greater (densities are estimated relative to the actin signal in muscles, Fig. 6D). Note that a slight anisotropy of the mouse lattice of placodes is detectable (SI Appendix, Fig. S9F) and could be caused by a small chemotactic or growth anisotropy in that species (SI Appendix, Fig. S10A). In addition, the stretch of the lattice is much easier to detect in the spiny mouse (Fig. 5F) not only because its amplitude is larger, but also because any anisotropic stretch is more readily visible on highly ordered lattices.

Taken together, our analyses demonstrate that the long-range patterns with spacing anisotropy in the spiny mouse are not generated by an expansion-induction mechanism, but might be dominated by a chemotaxis process with anisotropic random motility of dermal mesenchymal cells, anisotropic chemotaxis strength, and anisotropic growth. Our findings also highlight the broad applicability of the reaction–diffusion–chemotaxis framework in describing skin appendage patterns with different levels of spatial order.

Discussion

The expansion-induction model, introduced more than 10 y ago (10), is attractive because of both its simplicity and effectiveness in predicting the position and timing of the newly appearing follicles in the laboratory mouse embryo. However, this model is not compatible with current knowledge of the molecular and cellular mechanisms involved in placode formation, especially the combined existence (in mammals, birds, and reptiles) of reaction–diffusion interactions at the cellular level (11, 12, 21) and a mechano-chemical feedback of mesenchymal (dermal) cell chemotaxis (21–30). In particular, the latter requires FGF signaling from the epidermis. Indeed, the aggregation of dermal cells mechanically compresses the overlying epidermis, which responds by producing FGF that diffuses in the dermis. As dermal cells aggregate by chemotaxis along the FGF concentration gradient, this mechano-chemical response forms a positive feedback loop to placode formation.

Here, we show that the chemotactic component of the self-organized reaction–diffusion–chemotaxis system can, in isolation, give rise to the effective geometric rule that initially justified the expansion-induction model. Using numerical simulations, we find that placode insertion dynamics in the chemotaxis model under isotropic expansion robustly match to those observed in the laboratory mouse, demonstrating the equivalence of the geometric rules emerging from the chemotaxis and expansion-induction models. In addition, our mathematical analyses of a minimalistic 1D scenario show that this geometric rule might be a generic property of the chemotaxis model under isotropic expansion, i.e., it is robust to variations of the system parameters, provided that it gives rise to spot states. We additionally note that the presence of mesenchymal cell turnover kinetics—finite production p_m and degradation d_m

coefficients—is indispensable for this mechanism to emerge. Indeed, in a system with a conserved total cell density—i.e., with no production nor degradation—existing spots would simply “move apart” due to domain expansion, and no new spots would be formed. This points to the importance of cell turnover kinetics, which is, in some cases, neglected when describing mesenchymal cell dynamics in spatial patterning of placodes (22, 24, 25, 36, 37, 52).

While production and degradation represent a simple “source-sink” type of cell turnover kinetics in the chemotaxis model, we have not considered the effect of mesenchymal cell proliferation, which may play an important role in the formation of dermal condensates (53). In *SI Appendix, section 8*, we introduce a variation to the chemotaxis model, incorporating a proliferation-type of cell turnover, and study the dynamics of a nonlinear spot state under isotropic expansion. We find that the model with cell proliferation generates new spots through dynamics (*SI Appendix, Fig. S13*) highly similar to those in *Fig. 3* and *SI Appendix, Fig. S5*, suggesting that the spot insertion mechanism might be a generic property of chemotaxis models, regardless of the specific form of cell kinetics.

Note that the expansion-induction model was motivated, at least in part, by the observation that reaction–diffusion systems used to describe mouse follicle patterning (11, 12, 16) might not capture the spatio-temporal characteristics of this process. Our supplementary analyses (*SI Appendix, section 9*) show that a Gierer–Meinhardt model—a two-component RD model classically employed to describe skin appendage patterning (54, 55)—produces new spots in an expanding domain through the splitting of existing ones, even when primary spot positions are held fixed (*SI Appendix, Fig. S14*). However, analyses of other RD systems have shown that, depending on the reaction kinetics, either peak/spot splitting (56, 57) or insertion (58–60) dynamics may occur in one and two dimensions. Hence, whether any such system can quantitatively reproduce the patterning of laboratory mouse placodes, in the absence of a chemotactic component, remains an open question for future work.

From an experimental point of view, there is evidence that hair placode patterning is driven both by RD-like interactions at the epidermal level and chemotactic interactions at the epidermis–dermis interface, and that these interaction modes have separate patterning capabilities (21, 27). In this respect, we suggest here that the placode insertion process observed in mammals is dominated by the chemotactic component of the underlying self-organizational system.

We also investigate follicle patterning in the spiny mouse, which is characterized by a regular pattern with long-range order and anisotropic follicle spacing. While this pattern is incompatible with the expansion-induction model, the chemotaxis model can generate patterns closely resembling that of the spiny mouse if anisotropies in chemotactic strength and in the random motility of dermal mesenchymal cells are introduced. The similarity is remarkable because it is achieved simultaneously in multiple statistical aspects of the pattern: its long-range order (illustrated by the distributions of nearest-neighbor distances and their angles) and the elliptic shape of placodes (*Fig. 5*). On the other hand, the anisotropy in their spacing is caused by a combination of anisotropies in chemotactic strength and embryonic growth (*Fig. 5* and *SI Appendix, Fig. S10*).

Further studies are warranted to identify whether other factors might affect the in-vivo spatio-temporal evolution of the pattern.

In general terms, simple geometric rules emerging from underlying more intricate processes can be highly effective in predicting developmental processes. For example, a cellular automaton predicts the dynamics of monochromatic (green or black) skin scale color flipping (61–63), as well as a neighborhood-dependent color sub-clustering of scales (64) in multiple lizard species. In that case, the effective cellular-automaton behavior is not only confirmed in other species (65, 66), but is also shown to emerge from the superposition of skin geometry with the reaction–diffusion interactions among colored cells (61, 62). Conversely, although the geometric rule of the expansion-induction model is appealing, it does not generalize to multiple species and its assumptions—existing placodes acting as cues of positional information for the diffusion of an inhibitory signal—are not supported by current molecular developmental data. It remains to be shown if analyses in other species would confirm our proposition that the chemotactic term plays a central role in the process of reaction–diffusion–taxis patterning of skin appendages in mammals and possibly other vertebrate lineages.

Materials and Methods

Details on the experimentation model, whole-mount in situ hybridization, embryonic skin immunostaining and whole-mount imaging, actin filament and nuclei staining, as well as surface mesh and placode detection are provided in *SI Appendix, section 11*.

Numerical Model and Mathematical Analyses. The supporting information describes in details the continuum model for dermal chemotaxis, its linear stability analysis, its expression in co-deforming coordinates, the mathematical analysis of spot insertion dynamics, methods for the comparison of simulations results with experimental data, dynamics of spot formation in a pure reaction–diffusion model (i.e., without chemotaxis), details of our numerical implementation and description of statistical tests used in the main text. Simulations on 2D surfaces (*Figs. 2–4*) are based on finite difference methods, whereas simulations on 3D surfaces (*Fig. 5* and *SI Appendix, Fig. S9*) are based on finite element methods, with details provided in *SI Appendix, section 4*.

Data, Materials, and Software Availability. The executables and original code data have been deposited in public Code Ocean capsule (<https://codeocean.com/capsule/5330829/tree/v1>) (67). All other data are included in the manuscript and/or supporting information.

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